

Money and the Financial System

The financial system

- The financial system consists of different financial institutions
- The financial system includes markets, firms, and other institutions which carry out financial decisions of the households, businesses, and the government
- Financial system
 - Channels resources from savers to investors
 - Has become complicated over time (earlier banks; now security markets, pension, insurance)
 - Consequences of stability/failure of a financial system: growth/recession

Financial institutions

- Financial institutions can be grouped into two categories:
 - Financial markets: are the financial institutions that link savers directly to the borrowers
 - Bond market
 - Stock market
 - Financial intermediaries: are the financial institutions through which savers indirectly provide funds to borrowers
 - Banks
 - Mutual funds

The bond market (1)

- The Tatas re-entered the aviation sector a few years back. Suppose they want to increase their fleet size
 - They can borrow directly from the public by selling bonds
 - Bond: a certificate of indebtedness that specifies the obligations of the borrower to the holder of the bond
 - A bond identifies
 - Principal: the amount borrowed
 - Date of maturity: the time of repayment of the loan
 - Rate of interest to be paid until maturity
 - The buyer can hold the bond until maturity or can sell it at an earlier date

The bond market (2)

- In general, there are many bonds on offer. Even the government may borrow money using this mode of financing. Bonds differ with respect to the following characteristics
 - Term: length of time until the bond matures
 - Interest rate depends on the tenure
 - Generally, long-term bonds pay more. Why?
 - What if the holder needs the money earlier than the date of maturity?
 - In UK, a *perpetuity* bond – a bond that never matures (the principal is never paid) and pays interest forever – was also issued

The bond market (3)

– Credit risk

- is the probability that the borrower will default (i.e., will fail to pay some of the interest and principal)
- There is a trade off between risk and returns
- Government bonds are among the safest and pay lower interest
- Buyers can judge credit risk using the ratings given by the credit rating agencies

– Tax treatment

- The way the tax laws treat the interest earned on the bond
- The interest on most bonds is taxable
- One rationale for low interest rate on the government bonds is the tax advantage



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APPLICATION FORM

TRANCHE 1 ISSUE OPENS ON : WEDNESDAY DECEMBER 28, 2011
TRANCHE 1 ISSUE CLOSING ON : WEDNESDAY JANUARY 11, 2012^

Options	Series of Bonds**	
	Tranche 1 Series 1 (I)	Tranche 1 Series 2 (II)
Tenor	10 years	15 years
Redemption Date	At the end of 10 years from the Deemed Date of Allotment	At the end of 15 years from the Deemed Date of Allotment
Redemption Amount (₹ / Bond)	Repayment of the Face Value plus any interest that may have accrued at the Redemption Date	Repayment of the Face Value plus any interest that may have accrued at the Redemption Date
Frequency of Interest Payment	Annual	Annual
Minimum Application Size	₹ 50,000 (50 Bonds)	
In Multiples of	₹ 1,000 (1 Bond)	₹ 1,000 (1 Bond)
Face Value (₹ / Bond)	₹ 1,000.00	₹ 1,000.00
Issue Price (₹ / Bond) (A)	₹ 1,000.00	₹ 1,000.00
Mode of Interest Payment	Through various modes available*	Through various modes available*
Coupon Rate (%) p.a.	8.20% per annum	8.30% per annum

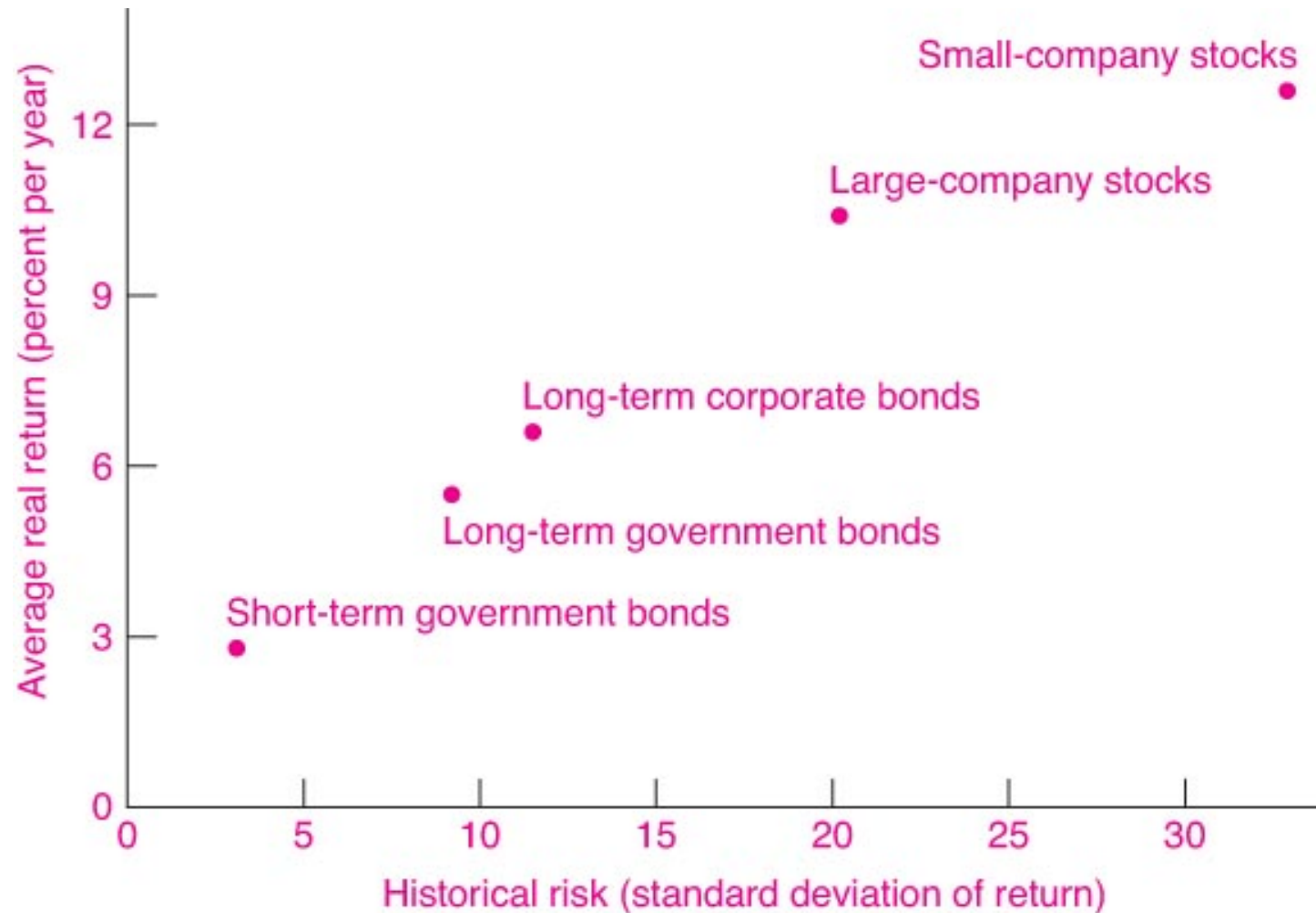
CREDIT RATING

“CRISIL AAA/Stable” by CRISIL, “CARE AAA” by CARE and “Fitch AAA(ind) with Stable Outlook” by FITCH

The stock market (1)

- The Tatas can also sell stock (in the new company)
- Stock represents ownership in a firm. Stock is a claim to partial ownership in a firm (and, therefore a claim to the profits accruing to the firm)
 - If Tatas sell 100 shares of stock, each share will represent the ownership of 1/100 of the business (the new company)
- Stock and bond financing
 - Stock: equity finance; bond: debt finance
 - The stockholders receive the benefits of the profits (dividend), if a company is profitable. The amount of dividend usually varies depending upon the firm's performance
 - The bondholders receive the interest at a constant rate. They would receive it even if the firm incurs losses (contrast it to stock – a loss-making unit may decide not to declare any dividend)
 - Financial difficulty: bondholders are paid their dues before stockholders
 - Compared to bonds, stocks generally offer higher return but also have higher risk

Risk and return trade-off



The stock market (2)

- Two types of investors
 - Impatient investors; they speculate on the future value of assets rather than wait patiently
 - Those who invest on firm foundations
 - Firm foundations: assets should be valued based on their intrinsic value
 - For common stocks, intrinsic value is the present value of the dividends
 - “slow but a safe way of getting rich”
- Speculative bubbles occur when prices rise because people think that prices are going to rise even further in the future (*a la* Keynes’ Animal Spirits)
 - Land prices at the time of boom
 - If people purchase stocks anticipating that the prices will rise in the future, this sends up the prices. The others follow the bandwagon

The stock market (3)

- History is marked by bubbles in which speculative trading drives the prices up beyond the intrinsic value of assets. The bubbles usually leads to crashes
 - Tulip Mania and the South Sea Company [cases](#)
 - Netherlands: Tulips as the currency of the future
 - 1710s: rising debt in the UK, the Company was granted monopoly to trade in South Seas (including trades of African slaves). The Company offered promises, people invested but the trade did not materialize. The King took over, it led to more confidence and investment as the Company offered abnormal returns. Soon, “the bubble began to wobble”: revenue was already low, the Company insiders engaged in malpractices and bribing to keep the price high, but it did not last long; even Isaac Newton lost “as much as £40 million of today’s money in the scheme”
 - Another famous bubble is of 1920s in the US
 - Then, credit (‘buy now, pay later’ practices) was introduced for the first time in the US. This offered an opportunity to a war savaged economy and led to large increases in aggregate spending. People took loans and bought stocks. Debt also increased substantially. The *Roaring Twenties* were followed by the Great Depression
 - Recent bubbles – the dotcom (2000s) and the housing bubble (~2007)

The stock market (4)

- Stock market index
 - Is the weighted average of the prices of a basket of company stocks
 - Helpful in tracking trends in stock markets
 - Indices are sensitive to domestic and international political and economic scenarios
 - BSE Sensex
 - 30 companies; representative of various sectors
 - Estd: 1875; base period: 1978–79; base value: 100
 - NSE Nifty 50
 - 50 companies
 - Estd: 1992; base period: Nov 1995; base value: 1000

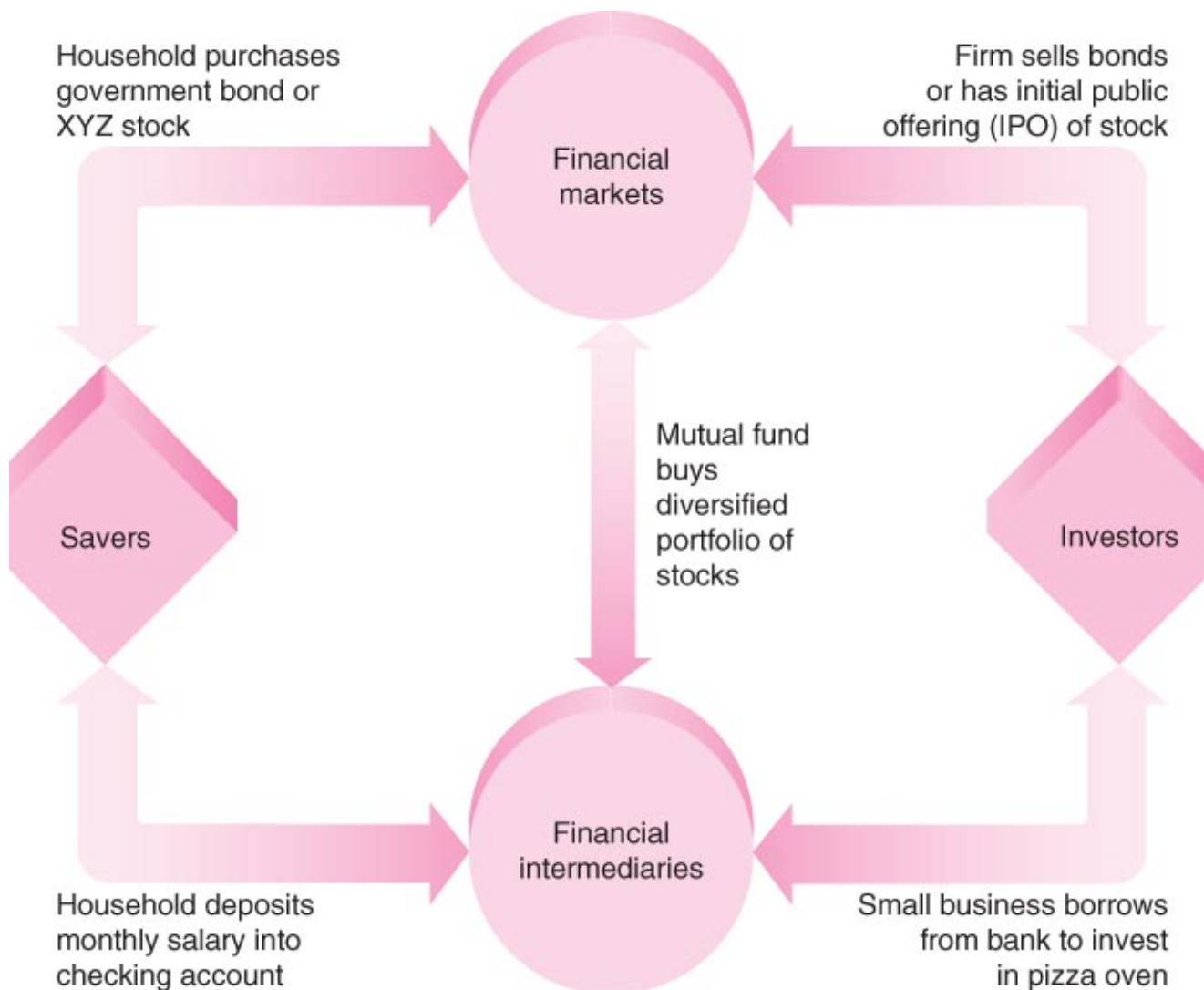
Banks

- A small grocery owner (unlike the Tatas) will find it difficult to raise funds in the stock and bond markets in order to expand his business
 - Will you be ready to buy stocks/ bonds floated by a small shop keeper?
- Banks accept deposits from people who want to save and use them to make loans to borrowers
 - Banks charge slightly higher interest rates from the borrowers
 - Besides being financial intermediaries, banks facilitate transactions in the economy through, say, *cheques*. Thus, banks create a special asset which acts as medium of exchange. (What about stocks and bonds?)
 - Bonds and stocks are not transferrable

Mutual Funds

- An institution that sells shares to the public and uses the proceeds to buy a portfolio of stocks and bonds
 - The shareholder of the mutual fund accepts all the risk and returns associated with it
 - Allow people with small amounts of money to diversify their portfolio
 - Mutual funds give ordinary people the access to skills of professional money managers, who keep a close eye on the market. The company operating the mutual fund charges a fee from the shareholders

Flow of funds



The financial system: functions

- Functions
 - Transfers resources across time, sectors, and regions
 - Intergenerational transfer
 - Households (savers) to Firms (investors)
 - Cross-country transfer too, from the countries with high savings rate to the ones having high investment opportunities
 - Risk management
 - Insurance
 - Pooling and subdivision of funds
 - Example, equity
 - Clearing house

Financial assets

- What are real assets?
- Financial assets are the financial claims by one party against another
 - Money
 - Savings accounts
 - Stocks
 - Pension funds
 - Credit market instruments (e.g., govt securities)
- Money
 - Is a commonly accepted medium of exchange
 - “Money is useless until we get rid of it”
 - In macroeconomics, this term is used in a specific sense and differs from wealth. Bill Gates, for instance, may not have lot of money though he may be among the wealthiest

Evolution of Money (1)

- Barter system: exchanging goods for other goods
 - Would not it complicate the matters by replacing one transaction with two?
 - What if the items received in barter trade are not acceptable
 - Difficulty in locating the person whose desires compliment your own
 - Other problems
 - Once a singer from Paris gave a concert in an island. In exchange, her payment consisted of 3 pigs, 23 turkeys, 44 chickens, 5000 cocoanuts and lot of fruits...so much that it would have brought her huge money. Since the money was scarce in the island, she was paid as above. And since she could not consume the receipts, it became necessary in the meantime to feed the pigs and poultry with the fruits!
 - As economies develop, people no longer engage in the barter trade. But they sell goods for money and then use money to buy other goods

How money simplifies exchange of commodities

- We are comparing the number of prices (exchange ratios) needed in a monetary economy to that in a barter economy
 - Assume there are 4 commodities in the economy – c_1 , c_2 , c_3 , and c_4
- In a barter economy, we need 6 prices
 - $c_1:c_2$, $c_1:c_3$, $c_1:c_4$, $c_2:c_3$, $c_2:c_4$, $c_3:c_4$
- In a monetary economy, 4 prices may suffice
 - $m_1:c_1$, $m_2:c_2$, $m_3:c_3$, $m_4:c_4$
- n commodities
 - Barter economy: $n*(n-1)/2$
 - Monetary economy: n

Cowries

(Long, J. 1973 (1869). *Selections from Unpublished Records of Government for the Years 1748 to 1767 Inclusive Relating Mainly to The Social Condition of Bengal*. Calcutta: Firma K L Mukhopadhyay)

Cowries are shells
of sea snails

Typical uses in
games, jewelry,
prayers, rituals,
etc.,

Their use as the
medium of
exchanges extend
beyond India to
China, Lanka,
Maldives, and
several countries
in Africa

Coinage And Cowries.

The original payments were made in cowries, which must have been a very cumbrous way of exchange. In 1753 a Captain Cook refused to take cowries on board, though the Court had directed that ten tons of cowries were to be laden on each of the ships homeward bound ; he alleged the loss by pilferage and the loss of weight when they were cleared.¹ In 1758 the heads of offices were to indent on the Buxey for cowries they may want. We find that as late as 1767 the revenues of Sylhet were paid in cowries.

One of the earliest efforts to remedy this deficiency in coin was to have a Mint in Calcutta, though the Nawab and Jagat Set opposed on account of the large profits they drew from their Mint at Murshidabad.²

Evolution of Money (2)

- Commodity money
 - Cattles, and items such as beer, copper, iron, silver, gold served as money over time
 - Disadvantages: Indivisibility, wastage, and handling
 - In the 18th century, commodity money was largely limited to metals. These have intrinsic values. Limitation: restricted supply (scarce resources were needed to dig it out)
 - It eventually led to a more stable currency system
- When an economy uses gold as money it is said to be operating under the **gold standard**
 - In the late 19th century international trade was conducted on the basis of the gold standard – gold was used to settle trade transactions between nations
 - Those with trade surplus accumulated gold whereas those with trade deficits witnessed fall in their gold reserves
- The gold standard existed till World War II and after the birth of IMF, it was largely abolished

A Chemist Explains Why Gold Beat Out Lithium, Osmium, Einsteinium ...

November 19, 2010 · 12:01 AM ET

Heard on [Morning Edition](#)

By [Jacob Goldstein](#), [David Kestenbaum](#)

Source: <https://www.npr.org/sections/money/2011/02/15/131430755/a-chemist-explains-why-gold-beat-out-lithium-osmium-einsteinium>

- The periodic table lists 118 different chemical elements. And yet, for thousands of years, humans have really, really liked one of them in particular: gold. Gold has been used as money for millennia, and its price has been going through the roof
- Why gold? Why not osmium, lithium, or ruthenium?
- We went to an expert to find out: Sanat Kumar, a chemical engineer at Columbia University. We asked him to take the periodic table and start eliminating anything that wouldn't work as money
- Sanat starts with the far-right column of the table. The elements there have a really appealing characteristic: They're not going to change. They're chemically stable
- But there's also a big drawback: They're gases. You could put all your gaseous money in a jar, but if you opened the jar, you'd be broken. So, Sanat crosses out the right-hand column
- Then he swings over to the far left-hand column, and points to one of the elements there: Lithium. "If you expose lithium to air, it will cause a huge fire that can burn through concrete walls," he says
- Money that spontaneously bursts into flames is clearly a bad idea. In fact, you don't want your money undergoing any kind of spontaneous chemical reactions. And it turns out that a lot of the elements in the periodic table are pretty reactive. Not all of them burst into flames. But sometimes they corrode, start to fall apart
- So, Sanat crosses out another 38 elements, because they're too reactive

- Then we ask him about those two weird rows at the bottom of the table. They're always broken out separately from the main table, and they have some great names — promethium, einsteinium. But it turns out they're radioactive — put some einsteinium in your pocket, and a year later, you'll be dead
- So we're down from 118 elements to 30, and we've come up with a list of three key requirements:
 - Not a gas
 - Doesn't corrode or burst into flames
 - Doesn't kill us
- Now Sanat adds a new requirement: You want the thing you pick to be rare. This lets him cross off a lot of the boxes near the top of the table, because the elements clustered there tend to be more abundant. At the same time, you don't want to pick an element that's too rare. So, osmium — which apparently comes to earth via meteorites — gets the axe
- That leaves us with just five elements: rhodium, palladium, silver, platinum and gold. And all of them, as it happens, are considered precious metals. But even here we can cross things out. Silver has been widely used as money, of course. But it's reactive — it tarnishes. So, Sanat says it's not the best choice
- Early civilizations couldn't have used rhodium or palladium, because they weren't discovered until the early 1800s. That leaves platinum and gold, both of which can be found in rivers and streams
- But if you were in the ancient world and wanted to make platinum coins, you would have needed some sort of magic furnace from the future. The melting point for platinum is over 3,000 degrees Fahrenheit. Gold happens to melt at a much lower temperature, which made it much easier for pre-industrial people to work with

Evolution of Money (3)

- Modern money
 - Paper money or fiat money
 - Wanted not for its own sake but for the things it can buy. We do not consume it directly
 - Convenience, storage
 - Future: cashless society?
- Legal tender and fiduciary money
 - Currency and coins serve as money by the order (fiat) of the govt (unlike commodity money, these do not have intrinsic value)
 - They are legal tenders and are accepted in settlement of payments of all kind
 - The same is not true of demand deposits though. The latter are accepted as money on trust and are not legal tenders. A payee can legally refuse to accept the payment via cheque (and there is no guarantee that it will be honoured at the bank)

Functions of money

- Medium of exchange
 - It is this function which separates money from other financial assets
 - Money's value as a medium of exchange is more obvious when there is any monetary disruption
 - 1990s: Russia abandoned its central planning system, people spent hours waiting for goods and to get dollars (the Ruble had ceased to function as an acceptable means of exchange)
 - 2016: Demonetisation
- Unit of account
 - It is the unit by which we measure value of things (goods and services)
 - National income estimates
- Store of value
 - Can we store perishable (e.g., vegetables) items for long periods?
 - People can hold their wealth in the form of money
 - In comparison to some risky assets, money is relatively riskless
 - Money is not perishable, and its storage costs are also considerably lower. However, in modern economies, people rarely hold money for this reason

Demonetisation

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Cost of holding money

- What is the cost of holding money?
 - It is costly to hold money since it has a lower yield than that of the other safe assets
 - I have Rs 100,000
 - I keep it in my safe for a year. After one year, I shall still have Rs 100,000. Currency has zero interest rate
 - I keep it in my savings account. After one year, I shall have Rs 104,000
 - In the above case, the cost of holding money is 4 per cent per year
 - The cost of holding money is the interest forgone from not holding other assets
 - Interest rate: is the price paid for borrowing money

The supply of money (1)

- Money supply is the total stock of money in circulation in public at a given point in time
- The money supply comprises of different components: currency, demand deposits, and time deposits
 - Currency (coins and fiat money) and demand deposits together are known as transactions money. These are used to settle transactions
 - Demand deposits is the amount of money held in banks or other financial institutions which you can withdraw on demand immediately (e.g., balance in savings and current accounts)
 - Is FD a demand deposit?
 - Time deposits usually have a fixed period to maturity

The supply of money (2)

- Various components of the money supply have been aggregated in descending order of liquidity
 - Liquidity: refers to the ease with which an asset can be converted into the economy's medium of exchange
 - Reserve Money (M0) = currency + Bank's deposits with RBI + Other deposits with RBI (e.g., deposits by quasi-govt bodies, foreign central banks, accounts of governments)
 - Narrow money (M1) = M0 + term deposits of shorter duration (< 1 year)
 - Broad money (M3) = M1 + term deposits of longer duration (> 1 year)
- By the term money supply, we usually mean M3. It is also referred to as **aggregate monetary resources**
- Generally, the size of M3 is much larger compared to M1
 - In India, M3 in 2010-11 was Rs 70,000 billion and M1 comprised nearly one-fifth of this amount

The demand for money (1)

- How does the demand for money differ from that of food?
 - As discussed, money is not demanded for its own sake
 - The demand for money is an **indirect** or **derived** demand. Money is demanded because of its functions (?)
- The demand for money is affected by the following two reasons
 - Transactions demand
 - People hold money to bridge the time gap between their income and expenditure. They receive income at discrete points in time and spend it continuously throughout an interval
 - As incomes rise, both income and expenditure streams grow and so is the transactions demand. For examples, if all prices and incomes double, the demand for money doubles
 - Is thus a positive function of the level of income, y

The demand for money (2)

- Speculative demand (or the demand for liquidity preference)
 - People have an option of keeping money in other assets. Let us club all such assets into bonds. The inclination to hold money also depends on the returns on bonds. This is known as speculative demand
 - Is negatively related to the interest rate, r
- Total demand = transactions demand + speculative demand
- $M_d = k(y) + l(r)$
- Exceptions
 - Sometimes, money may also be demanded for other reasons. In countries with hyperinflation, for example, the USD is demanded. Such demand relates to the store of value function of money, and such instances are exceptions

Do credit cards affect the money demand?

Why Credit Cards Aren't Money



It might seem natural to include credit cards as part of the economy's stock of money. After all, people use credit cards to make many of their purchases. Aren't credit cards, therefore, a medium of exchange?

At first this argument may seem persuasive, but credit cards are excluded from all measures of the quantity of money. The reason is that credit cards are not really a method of payment but a method of *deferring* payment. When you buy a meal with a credit card, the bank that issued the card pays the restaurant what it is due. At a later date, you will have to repay the bank (perhaps with interest). When the time comes to pay your credit card bill, you will probably do so by writing a check against your checking account. The balance in this checking account is part of the economy's stock of money.

Notice that credit cards are very different from debit cards, which automatically withdraw funds from a bank account to pay

for items bought. Rather than allowing the user to postpone payment for a purchase, a debit card allows the user immediate access to deposits in a bank account. In this sense, a debit card is more similar to a check than to a credit card. The account balances that lie behind debit cards are included in measures of the quantity of money.

Even though credit cards are not considered a form of money, they are nonetheless important for analyzing the monetary system. People who have credit cards can pay many of their bills together at the end of the month, rather than sporadically as they make purchases. As a result, people who have credit cards probably hold less money on average than people who do not have credit cards. Thus, the introduction and increased popularity of credit cards may reduce the amount of money that people choose to hold.

**Slides beyond this won't be a part of
the syllabus for the end-term exam**

The central bank (monetary authority)

- Plays an important role in formulation of macroeconomic policy
 - Actions influence not just the domestic economy but also the economy of other countries
- Goals
 - Generally, central banks have multiple goals
 - Economic stability
 - Inflation
 - Unemployment
 - Exchange rate

Some functions of the central bank

- Monetary policy
 - Formulation and execution of monetary policy
 - Manages short-term interest rates
 - Issuing currency and its management
- Banker and debt manager to Government
 - Banker to central and state governments
- Banking institutions
 - Supervision and regulation
 - Licensing
 - Enforcing various norms
 - Provides financial services
 - Helps in settling inter-bank transactions
 - Maintaining bank reserves
 - Lender of the last resort
 - Banks facing temporary liquidity crisis
- Financial system
 - Ensures stability

Balance sheets

Balance sheet: a statement of a firm's financial position at a time;

Assets: items that a firm owns; *Liabilities*: items that a firm owes

Commercial Banks	
<i>Liabilities</i>	<i>Assets</i>
Demand deposits	Reserves
Time deposits	Loans/ advances
Other liabilities	Investments and securities
Central Bank	
<i>Liabilities</i>	<i>Assets</i>
Currency	Govt securities
Bank reserves	Loans and other credits
Other deposits and liabilities	Other assets

Some definitions

- High powered money
 - Is the sum of the liability of the central bank
 - Currency in circulation
 - Reserves of commercial banks deposited with RBI
 - The central bank may acquire assets against these liabilities
- Bank rate
 - Is the interest rate at which RBI lends money to commercial banks
- Reserve deposit ratio
 - Is the share of deposits that commercial banks keep as reserves
 - *Cash Reserve Ratio (CRR)*: is the share of deposits that commercial banks keep with RBI
 - *Statutory Liquidity Ratio (SLR)*: is the share of deposits that commercial banks keep with them in safe and liquid assets like cash, gold, and government securities

Monetary policy instruments (1)

- Open market operations
 - Central banks affect the money supply in economy by selling and buying government securities in the open market
 - Suppose RBI sells government securities
 - It sucks out currency from the public
 - The money supply (high powered money) in the economy decreases

Monetary policy instruments (2)

- Reserve requirements
 - Banks do not keep the entire deposits as reserves
 - Seldom the entire amount would be withdrawn together
 - Such reserves in banks' vaults earn no interest, why not go for investment
 - But, in many cases, the banks had insufficient funds to meet depositor's demand
 - As a regulator, the central banks fix reserve requirements
 - When Cash Reserve Ratio is high
 - Banks need to keep more money as reserve, it reduces the money supply in an economy

Monetary policy instruments (3)

- Bank rate
 - The banks borrow money from RBI if they fall short of reserves
 - Low bank rate: borrowing is cheaper; banks lend out more loans; increase in the money supply
 - High bank rate: borrowing is costlier; banks are discouraged to borrow more; reduction in the money supply
 - In case of high inflation, the central bank may increase the bank rate

Monetary policy instruments (4)

- Sterilisation

- To stabilise the economy vis-à-vis external shocks
 - Better growth prospects in India: investors abroad want to buy Indian bonds
 - They need to convert foreign currency to Indian Rupee
 - A commercial bank will convert. It will deposit the foreign currency with RBI and get the Indian currency
 - RBI: forex holding goes up, the money supply in the economy has increased
 - RBI will undertake open market operations (sell?)

Monetary transmission mechanism (1)

- Monetary transmission mechanism
 - Is the process through which the monetary policy (various actions undertaken by the monetary authority) interacts with banks and the rest of the economy
- How do the variations in money supply affect output, employment, prices, and inflation?
 - Quantity theory of money
 - Keynesian approach

Monetary transmission mechanism (2)

- Quantity Theory of Money (QTM)
 - Changes in the quantity of money lead to proportionate changes in the price level
 - $MV=PQ$; M-money supply, V-velocity of money, defined as the average number of times per year money changes hand, P-price level, Q-output
 - A stable money supply would produce stable prices. QTM helps explain why the countries with low (rapid) money growth have moderate (high) inflation

Monetary transmission mechanism (3)

- Keynes' monetary theory
 - Keynesian theory argues that changes in the demand or supply of money affect the economic activity not directly (as in QTM) but indirectly via changes in the interest rate and investment
 - Money demand and supply – r – Investment – output – employment – prices and cost of production
 - Monetary policy in AS-AD framework: consider a reduction in r – higher I , C – higher AD – higher equilibrium Q and P (given AS)